Northeastern University
Bouvé College *of* Health Sciences

Course number: PHTH7440

Course name: Advanced Biostatistics

Instructor

Instructor: Sara Lopez Pintado

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Class Time and office hours:

Fall 2026: Thursday: 1:35pm – 4.05pm

Class Location: Hastings Suite 111

Credit hours: 3 semester hours

Office hours: TBD and by appointment

Course Information

Description:

Biostatistics is essential to ensuring that findings and practices in public health and biomedicine are supported by reliable evidence. Biostatistics is a field that applies statistical methodologies and reasoning to medical, biological, and public health problems. Advanced Methods in Biostatistics is a graduate course that extends general and generalized linear models to applications and structures that create dependent data. The course uses longitudinal and multilevel approaches to model clustered or group data that can arise from data collected longitudinally across time or at different layers. In the first part of the course, we will review linear models and generalized linear models. In the core part of the course, we will introduce methods to analyze longitudinal/dependent data and hierarchical data. An introduction to functional data is also provided by the end of the course. Some of the specific topics covered in this course are generalized regression, mixed effect and marginal models for analyzing longitudinal data, and an introduction to functional data analysis. The course will explore the underlying principles for these approaches and specific statistical methods for analyses. The topics are motivated by applications and real data sets. We will use R and R-Studio to analyze various datasets. Most of the homework assignments will have to be done using RStudio. We will have several guest speakers over the period of the course and students' presentations.

Prerequisites:

PHTH 6210 Applied Regression or equivalent course on regression.

Credit Hours: 3 semester hours

Objectives / Outcomes

The course seeks to provide participants with a solid understanding of how longitudinal, functional and multilevel models are applied in public and population health research, and how to use these models in their own research. Linear and generalized regression models are reviewed and extended to dependent data, more concretely, longitudinal data. Students will learn how to formulate longitudinal models in statistical terms, how they are related to other types of statistical models, and their relative benefits and limitations. Exploratory methods for functional data analysis will be introduced. R will be used to analyze real data sets.

Teaching Methods / Learning Experiences

Although there is some weekly variation, generally classes will consist of following sections.

1. Lecture: presentation of material and examples
2. Discussion of any homework due that day, readings or quizzes from the prior week
3. In-class group activities (some weeks presentation by students)
4. Computer lab/assignment in R and quizzes
5. There will a few guests' lectures

This course runs for a total of 14/15 weeks beginning September 10, 2026 and we meet once a week. Lessons are organized around specific course topics, and contain lecture slides, labs/assignments, and sometimes readings and quizzes. You can access these materials, complete them and upload them in Canvas (https://canvas.northeastern.edu). See more details below.

Computer Software:

We will use RStudio, an integrated development environment for R that includes a console, a syntax editor, and tools for plotting and debugging. Both R and R Studio are free open-source and run on the Windows, Mac, and Linux desktop.

The program can be downloaded at https://rstudio.com/products/rstudio/download/

We will use RStudio statistical software packages to explore and analyze data for the course.

Grading / Evaluation

Assignments (50% of course grade)

Assignments are usually due prior to the beginning of class on the day assigned. They should be uploaded to the Assignments section of Canvas. Although you may work with other students on homework assignments, your handed-in assignments must represent your own work. Note that we will often discuss the assignments in class the day they are due, and thus late assignments are not accepted. However, I fully understand that situations may arise that could prevent completion of an assignment. Please contact the instructor in advance if you will be unable to complete the assignment on time. The solutions of assignments will be presented in class.

Quizzes (20% of course grade)

Quizzes will sometimes be open-book and open internet format, but not always. Calculators and laptops are allowed.

Final Project (20% of course grade) Students will complete a "real world" data analysis project and write-up. Additional details on the project and format will be provided later in the semester. We will use the last weeks of class to present the project.

Participation (10% of course grade) This is a participatory course, and each student is expected to be an active participant in the work that occurs. Students are expected to have read any assigned readings or homework and to come to class prepared to discuss materials. The students will present some of their work in class.

Grading letter grades

93-100% = A

90-92% = A-

87-89% = B+

83-86% = B

80-82% = B-

77-79% = C+

73-76% = C

70-72% = C-

67-69% = D+

63-66% = D

60-62% = D- 59%
and below = F

The grades will be rounded to the nearest tenth (e.g 79.6%=80%)

*See University Grading Scales for UG and Grad here:
https://registrar.northeastern.edu/article/university-grading-system/

Materials / Resources

Text

(Strongly recommended) Applied Longitudinal Analysis (Second Edition). Garret M. Fitzmaurice, Nan M. Laird, James H. Ware, Wiley. 2011

(Optional) Multilevel Modeling Using R (Third addition). W Holmes Finch, Jocelyn E. Bolin. Chapman & Hall, 2024.

(Optional) Introductory Statistics with R. Dalgaard. 2nd Edition.

Canvas

Course information is available on CANVAS (https://canvas.northeastern.edu).

Class topics and slides, pre-class readings, projects, data, assignments, will all be posted on the site.

Course Policies

Student Expectations:

Attend all classes. Read and complete assigned materials prior to class and actively participate in class discussions and any online forums. Demonstrate an understanding of the use of statistics on assignments, quizzes, and projects. Demonstrate the ability to use statistical methods, modeling and programming in analyzing data, in particular longitudinal or multilevel data. Demonstrate the ability to clearly interpret, explain and present analyses from these models.

Class Behavior and Participation:

Class participation is expected and will be considered for the final grade. This is a graduate course and collaboration and discussion among students is encouraged. Attending class and arriving to class promptly is expected of each student (and shows respect for fellow students). Students will be called on in class and everyone is expected to be respectful of fellow classmates. Computers may be used during class only for the purpose of taking notes and working on projects/assignments in class. Cell phones should, at a minimum, be on silent, but preferably turned off. Non-adherence to these expectations of professional behavior will result in dismissal from class. Consistent with University policy, students must seek the course instructor's permission to record class lectures. We will have guest lectures during the course.

Make Up and Late Work Policy:

Please note that all due dates and times will be specified in Canvas according to the Eastern Standard Time zone (EST); plan to complete and submit all assignments accordingly.

Academic Integrity:

As health professionals and scientists-in-training, the highest level of ethical behavior is expected of all students. Plagiarism, cheating, and any form of unauthorized collaboration will not be tolerated and will be handled in accordance with University policies including the Academic Honesty and Integrity Policy. Requirements can be found at:
https://osccr.sites.northeastern.edu/academic-integrity-policy/

https://osccr.sites.northeastern.edu/code-of-student-conduct/

Use of Generative Artificial Intelligence (AI)

Generative Artificial Intelligence (AI) refers to large learning models (LLMs) that can generate high-quality text, images, and other content based on the data they were trained on. Popular AI "bots" include ChatGPT which produces text-based responses in the form of poems, essays, articles, letters and more based on a question entered by the user. It also can provide translation and copy language style and structure. LLMs are trained to predict the most relevant sequence of words in response to a prompt. AI tools are only able to draw on the information that was input, meaning their responses reflect the biases and limitations of the material they have been trained on. **It is important to understand that responses produced by generative AI tools will likely reflect consensus beliefs, including any biases and inaccuracies that inform those beliefs.**

In general, students will be able to use AI in this course unless it is specified otherwise. Please add references mentioning the use of AI. For in class quizzes, presentations and discussion students won't be able to use AI. Maintaining academic integrity is paramount: any content assisted or generated by AI must be properly attributed, and you must specify the role of the AI, the tool employed, and any input prompts at the end of an assignment. If a generative AI tool (i.e., ChatGPT) is used, your should acknowledge how it was used, even if no generative AI content was incorporated in the work. This acknowledgement should include:

· Which AI tool was used

· Describe how the AI tool was used

· Indicate the date AI tool was accessed

Students who use AI tools in the writing of an an assignment (including any part thereof), the production of images or graphical elements, or in the collection and analysis of data must be appropriately attributed and may not be submitted as if it were just the student's own work.

Belonging at Bouve' College:

We acknowledge that chronic and persistent disparities in education and health due to systemic oppression, based on race, class, and other marginalized identities, have adversely affected individuals and society. At Bouve´ we are committed to educating future healthcare leaders who strive to serve marginalized communities and improve the well-being of all members of society through the health span.

Within the college, we strive to create a safe environment for diverse thoughts, perspectives, and experiences to be shared, explored, and embraced. As a community we recognize that disrupting long-standing oppressive practices is a process. Everyone is at a different place on this journey, and we affirm our commitment to support and learn from one another.

In order for everyone in our learning community to grow, gain a deeper understanding of complex issues and be exposed to diversity of thought, we may have discussions in this class on topics that could be uncomfortable or difficult for some people. It is imperative that we follow rules of engagement as below in order to create a psychologically safe learning environment.

Ground Rules:

1. Listen actively and respect others when they are talking.
2. Participate to the fullest of your ability without taking up too much space – community growth depends on the inclusion of every individual voice
3. Speak from your own experience instead of generalizing ("I" instead of "we")
4. Do not be afraid to respectfully challenge one another by asking questions but refrain from personal attacks and stay focused on ideas rather than individuals
5. Be conscious of body language and nonverbal responses; they can be as disrespectful and hurtful as words.

Finally, all community members need to adhere to the NU Student Code of Conduct "e*ach Northeastern student has the responsibility to respect the differences of individuals and treat*

others in a civil and respectful manner." All work interactions in the class are expected to be courteous and respectful of other's health and cultural beliefs and opinions. Additional information about Northeastern's strong belief on fostering a community of belonging can be found online here:

Belonging at Northeastern

Students with Disabilities:

Disability Access Services (DAS), located on campus in 10 Dodge Hall (617-373-2675) can provide students with information and other assistance to help manage any challenges that may affect their performance in the coursework. The University requires that students provide documentation of their disability to the DAS. Students should meet with the course instructor for special accommodations to be arranged.

Northeastern University abides by Section 504 of the Rehabilitation Act of 1973, which stipulates that no student shall be denied the benefits of an education "solely by reason of a handicap." Disabilities covered by law include but are not limited to, learning disabilities and hearing, sight or mobility impairments. Additional information about DAS can be found online at https://drc.sites.northeastern.edu/

Student Wellness:

As health professionals (or research leaders) we do our best to maintain a healthy, balanced life. As a student however, you may experience a range of challenges including significant stress, difficult life events, mood changes, excessive worry, or problems with eating and/or sleeping. These can diminish your academic performance and/or reduce your ability to participate in daily activities. If you or anyone you know is struggling, we strongly encourage you to seek support. Northeastern University provides several services and resources to support the overall wellness of students:

University Health and Counseling Services

Find at Northeastern

We Care

Wellness Days

As always, please contact Northeastern University Police Department at 617-373-3333 in emergency situations.

Title IX Policy:

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including students, faculty and staff of all gender identities.

Learn more at:

https://www.northeastern.edu/ouec/training-and-education/syllabus-language-title-ix/

Know Your Options: Pregnancy

Course Feedback:

The Office of the Registrar, in collaboration with the Office of the Provost, administers the TRACE (Teacher Rating and Course Evaluation) for undergraduate and graduate courses at Northeastern University. Students' assessments through TRACE are completely anonymous. Given the public nature of student comments made through TRACE and their intended use to enhance teaching, students are asked to please be thoughtful, professional and considerate with responses.

Syllabus Disclaimer:

Any changes to the syllabus will be made at the discretion of the instructor. It is the responsibility of the student to stay informed of any such changes.

Tentative Course Schedule / Major Assignment

Week 1: September 10: Introduction to PHTH7440. Introductions. Go over Syllabus and overall course topics and organization. Linear Regression Model. Module 1.

Week 2: September 17. Continuation Linear Regression Model and Multivariate regression model. Review Linear Regression. Lab #1: Regression due on September 22 at 6pm.

Week 3: September 24. Generalized Linear Models. Introduction to Repeated Measures and Longitudinal Data. Repeated Measures ANOVA.

Week 4: October 1. Continuation of Generalized Linear Models. Introduction to Longitudinal data analysis and representation.

Week 5: October 8. Longitudinal data representation. Repeated Measures ANOVA (traditional very restrictive approach). Lab #2: Generalized Regression Models Review due October 8 by 11am.

Week 6: October 15. Mean Response Profile for longitudinal data analysis (when balanced data and categorical predictors).

Week 7: October 22. Modelling the mean with parametric and semiparametric functions (linear, quadratic and splines). Using GLS functions with unstructured covariance. Lab #3: Poisson Regression and Repeated Measures Anova due on October 22 by 11am.

Week 8: October 29. Modelling the covariance and Introduction to Mixed Effect models. Lab #4: Mean response profile analysis and parametric curves approach.

Week 9: November 5. Introduction to Mixed Effect Models Guest lecture from Prof. Javier Alvarez Liebana. Lab # 5: More on parametric curves and splines modeling of the mean due on November 5 by 11am.

Week 10: November 12. Review of mixed models (random intercept and random slopes models). Using lme function in R.

Week 11: November 19. Introduction to marginal models and GEE approach for non-continuous outcomes/Functional Data Analysis introduction. Guest lecture from Dr. Ally Hinton. Lab # 6: Mixed effect models, random intercept and slopes approach for modelling longitudinal data due November 19 by 11am.

Week 12: November 26. Thanksgiving Holiday

Week 13: December 3. Introduction to Functional Data Analysis. Non-parametrics methods. Final projects presentations 1.

Week 14: December 10. Final projects presentation 2